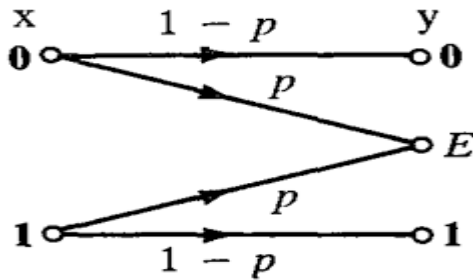
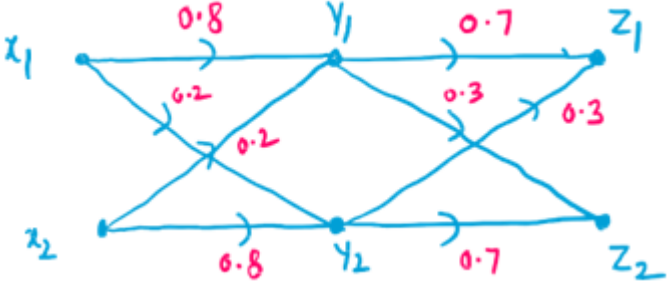


U23CC483 – Information Theory and Coding

(Regulations 2023)

CO1 :	Analyze coding techniques and discrete memoryless channels to optimize information transmission.
CO2 :	Apply error control coding techniques to detect and correct errors in digital communication systems.
CO3 :	Implement audio compression algorithms for efficient speech encoding and decoding while minimizing quality loss.

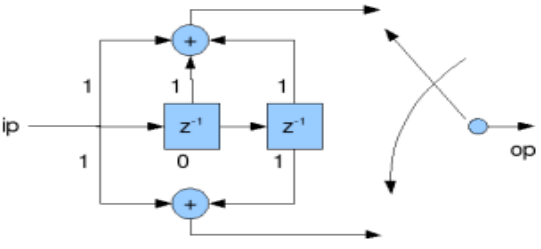
Q. No.		Question		M	CO#	KL
1	a)	(i)	Interpret the formula for information rate and explain what each term represents.	2	CO1	K3
		(ii)	Construct the Shannon-Fano code for a source with the following symbols and probabilities: {A: 0.35, B: 0.25, C: 0.20, D: 0.10, E: 0.10}. Calculate the average code length and compare it with the entropy of the source. Find Efficiency of the code.	14	CO1	K3
		(iii)	For the BEC of figure shown. Calculate the probabilities associated with the channel outputs for $p=0.2$, if the source outputs are equally likely. 	8	CO1	K3
1	b)	(i)	How is channel capacity affected by the noise in a Binary Erasure Channel (BEC)?	2	CO1	K3
		(ii)	Consider two random variables X and Y with the following joint probability distribution: $P(X, Y) = \begin{cases} 0.1 & \text{if } X = 0, Y = 0 \\ 0.4 & \text{if } X = 0, Y = 1 \\ 0.2 & \text{if } X = 1, Y = 0 \\ 0.3 & \text{if } X = 1, Y = 1 \end{cases}$	14	CO1	K3

			Calculate the joint entropy $H(X,Y)$ the marginal entropies $H(X)$ and $H(Y)$ and the mutual information $I(X;Y)$			
		(iii)	Two BSC's are connected in cascade as shown in the figure  (i) Find the channel Matrix of resultant channel Find $P(z_1)$ and $P(z_2)$ if $P(x_1)=0.6$ and $P(x_2)=0.4$	8	CO1	K3
2	a)	(i)	Identify the significance of Hamming weight and list its role in error detection.	2	CO2	K3
		(ii)	The parity check matrix of a particular (7,4) linear block code is given by $H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$ (i) Find the Generator Matrix G (ii) List all the code vectors (iii) What is the minimum distance between code vectors?	12	CO2	K3
		(iii)	Determine the encoded message for the following 8-bit data codes using the following CRC generating polynomial $P(x)=x^4+x^3+x^0$ i) 11001100 ii) 01011111	10	CO2	K3
2	b)	(i)	If a linear block code has a generator matrix G of size 3×5 , how many message bits can be encoded, and what is the length of the resulting codeword?	2	CO2	K3
		(ii)	The generator matrix for a (6,3) block code is given below. Find all the code vectors of this code. $G = \begin{bmatrix} 1 & 0 & 0 & : & 0 & 1 & 1 \\ 0 & 1 & 0 & : & 1 & 0 & 1 \\ 0 & 0 & 1 & : & 1 & 1 & 0 \end{bmatrix}$	10	CO2	K3
		(iii)	A rate 1/3 convolution encoder has generating vectors as $g_1=(100)$, $g_2=(111)$ and $g_3=(101)$	12	CO2	K3

			i) Sketch the encoder configuration. ii) Draw the codetree, state diagram and trellis diagram. iii) If input message sequence is 101110, determine the output of the encoder.			
3	a)	(i)	Interpret the purpose of companding in audio compression systems. Why is it important for improving dynamic range?	2	CO3	K3
		(ii)	Explain the working principles of waveform codecs, Source Codecs and Hybrid Codecs.	10	CO3	K3
3	b)	(i)	How would you apply frequency domain filtering to remove unwanted noise from an audio signal? Provide a basic explanation.	2	CO3	K3
		(ii)	Compare μ -Law and A-Law companding techniques in terms of their impact on signal-to-noise ratio (SNR) and dynamic range. In which scenarios would you choose one technique over the other?	10	CO3	K3

4	a)	(i)	List the significance of entropy in measuring the uncertainty of a source	2	CO1	K2																
		(ii)	A DMS has following alphabet with probability of occurrence as shown below: <table border="1"><tr><td>Sym</td><td>S0</td><td>S1</td><td>S2</td><td>S3</td><td>S4</td><td>S5</td><td>S6</td></tr><tr><td>P(x)</td><td>0.125</td><td>0.0625</td><td>0.25</td><td>0.0625</td><td>0.125</td><td>0.125</td><td>0.25</td></tr></table> Generate the Huffman Tree, Calculate the Entropy, Average Code word length and Efficiency of the above Probabilities.	Sym	S0	S1	S2	S3	S4	S5	S6	P(x)	0.125	0.0625	0.25	0.0625	0.125	0.125	0.25	12	CO1	K3
Sym	S0	S1	S2	S3	S4	S5	S6															
P(x)	0.125	0.0625	0.25	0.0625	0.125	0.125	0.25															
		(iii)	Describe the Source Coding Theorem. How does Shannon-Fano coding work? Illustrate with an example of constructing a Shannon-Fano code.	10	CO1	K3																
4	b)	(i)	Difference between a Binary Symmetric Channel (BSC) and a Binary Erasure Channel (BEC).	2	CO1	K2																
		(ii)	Consider a DMS with six possible symbols x_i , $i=1,2,\dots,6$ and the corresponding probabilities $P(x_1)=0.3, P(x_2)=0.25, P(x_3)=0.2, P(x_4)=0.12, P(x_5)=0.08, P(x_6)=0.05$. Find Entropy of the source, Average Code word length and Efficiency	12	CO1	K3																

		(iii)	Explain the concepts of joint and conditional entropies. How the entropies are used in determining the mutual information between two random variables? Give relevant examples.	10	CO1	K3
5	a)	(i)	Apply the concept of CRC to a specific data sequence, given a generator polynomial. How would you calculate the CRC for the data "1101011" using the polynomial "1011"?	2	CO2	K2
		(ii)	Analyze the process of generating codewords using a linear block code. Given the generator matrix G: $G = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$ a)Encode the message vectors 00, 01, 10, and 11 using the generator matrix. b)Discuss how the generated codewords form a linear block code. c)Explain the significance of the generator matrix in linear block codes.	10	CO2	K3
		(iii)	Consider the generation of a (7,4) cyclic code by generator polynomial $g(x)=1+x+x^3$. i)Calculate the codeword for the message sequence 1001 and construct systematic generator with G ii)Draw the diagram syndrome calculator generator by polynomial $g(x)$.	12	CO2	K3
5	b)	(i)	Show the differences between Turbo codes and convolutional codes in terms of error correction performance, decoding complexity, and real-world application. What are the trade-offs between the two coding schemes?	2	CO2	K2
		(ii)	The generator matrix for a (6,3) block code is given below. Find all the code vectors of this code. $G = \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right]$	10	CO2	K3
		(iii)	For the convolutional encoder shown in the figure, Determine the following:	12	CO2	K3

			 i)Dimension of the code ii)Code rate iii)Constraint Length iv)Generating sequences v)Output sequence for message sequence of $m=[1\ 0\ 0\ 1\ 1]$			
6	a)	(i)	How do source codecs differ from waveform codecs in terms of their approach to speech compression?	2	CO3	K2
		(ii)	A voice communication application requires high speech quality at a moderate bit rate. Apply the characteristics of waveform, source, and hybrid codecs to select the most suitable codec and justify your choice.	10	CO3	K3
6	b)	(i)	Difference between μ -Law and A-Law companding techniques used in audio compression.	2	CO3	K2
		(ii)	Apply suitable audio compression and speech compression techniques to design an efficient digital speech communication system capable of reducing bandwidth while maintaining acceptable speech quality. Justify your design choices.	10	CO3	K3

